

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Air Handling Clean Care Kit, Part Number 4919588, or equivalent
- Air Handling Clean Care Kit, Part Number 4919425, or equivalent

Additional Service Items

- No additional service items required.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Note : Brush away any loose dirt from around the area of the air handling connections to avoid contamination of the interior of the engine.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the air intake pipe to the turbocharger. See equipment manufacturer service information. Use protective caps from the Air Handling Clean Care Kit, Part Number 4919425, to cover the connection points.
- Remove the exhaust gas pressure tube. Refer to Procedure 011-027 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-027.html>)
- Loosen the V-band clamp on the inlet end of the exhaust gas recirculation (EGR) cooler.
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-007.html>)

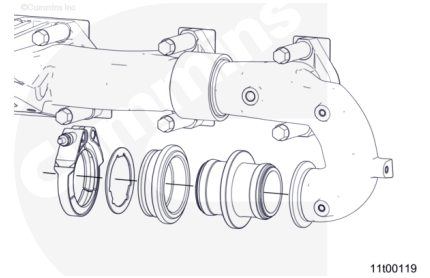
Remove

Remove and discard the gasket between the EGR cooler and the exhaust outlet pipe assembly EGR cooler connection.

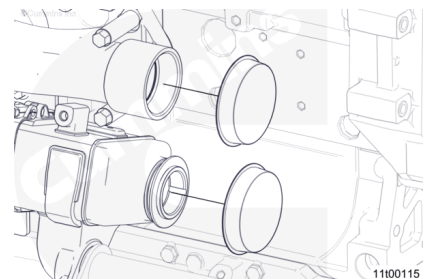
Remove the exhaust outlet pipe from the exhaust manifold end section by carefully tapping with a small dead blow hammer around the flange diameter, in order to avoid damage to the J-seals.

Do **not** remove the J-seals from the exhaust outlet pipe.

Confirm the exhaust outlet pipe has the J-seals on it. If **not**, check the bores for the J-seal.



Use protective caps from the Air Handling Clean Care Kit, Part Number 4919588, to cover open points in the air handling system.



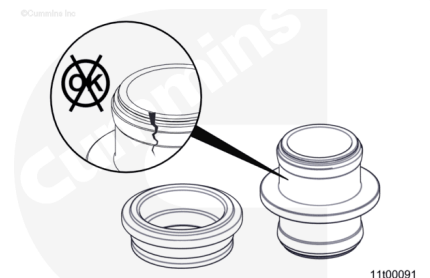
Clean and Inspect for Reuse

Clean the exhaust outlet pipe assembly EGR cooler connection with safety solvent.

Dry with compressed air.

Inspect the exhaust outlet pipe assembly EGR cooler connector for cracks or other damage.

Replace the exhaust outlet pipe assembly EGR cooler connection if damage is found.

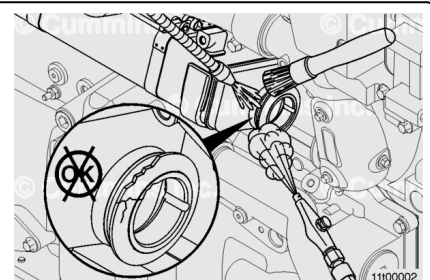


Clean the EGR cooler mounting flange with safety solvent.

Dry with compressed air.

Inspect the EGR cooler mounting flange for cracks or other damage. Replace the EGR cooler if cracks or other damage is found. Refer to Procedure 011-019 in Section 11.

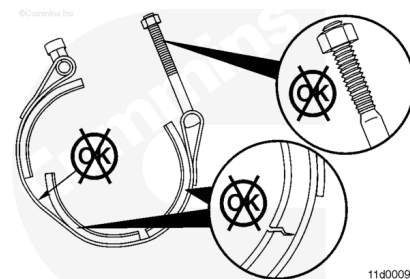
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Inspect the V-band clamps for signs of over-extension. The band **must not** be bent or cracked.

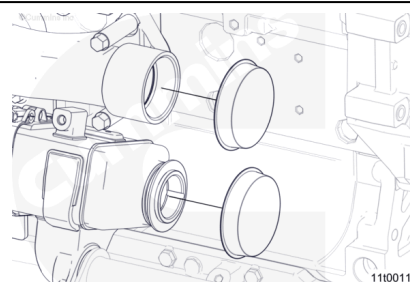
Inspect the V-band clamp threads for damage.

Replace the V-band clamp if damage is found.

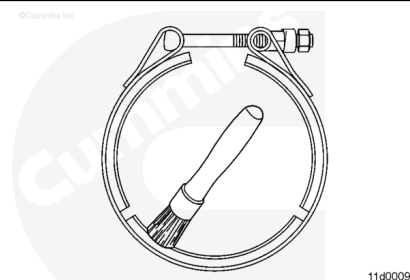


Install

Remove the protective caps from the Air Handling Clean Care Kit, Part Number 4919588, used to cover open points in the air handling system.



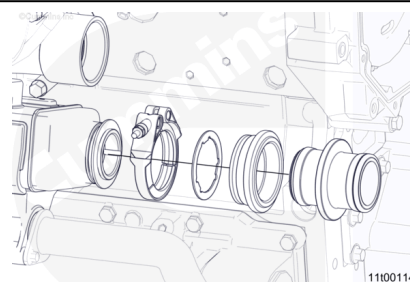
Before installing the V-band clamp, coat the clamping surface with high temperature anti-seize compound.



If separated, install the exhaust outlet pipe assembly EGR cooler connection to the EGR cooler adapter.

Install new gaskets on the pilot end of the exhaust outlet pipe assembly EGR cooler connection.

Install the V-band clamps loosely on the EGR cooler adapter.



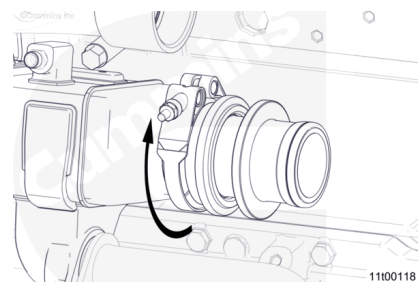
Install the exhaust outlet pipe assembly EGR cooler connection.

Position the V-band clamps that hold the exhaust outlet pipe assembly EGR cooler connection between the EGR cooler and the exhaust manifold adapter, as illustrated.

Rotate the exhaust manifold V-band clamp into position, as illustrated, and tighten the V-band clamps.

Note : Make sure the V-band joint is pushed together before installing and tightening the V-band clamp.

Torque Value: 17 n•m [150 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-007.html>)
- Install the exhaust gas pressure tube. Refer to Procedure 011-027 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-027.html>)
- Remove the protective caps from the intake piping.
- Install the air intake pipe to the turbocharger. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature. Check for leaks and proper operation.